



**SOLAR POWER GENERATION
MONITORING AND FAULT
DETECTION USING IoT**



A PROJECT REPORT

Submitted by

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In partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

**KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM
(An Autonomous Institution)**

ANNA UNIVERSITY : CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report **“SOLAR POWER GENERATION MONITORING AND FAULT DETECTION USING IoT”** is the bonafide work of **“MANIKANDAN K, MOHAN G, NAVEENKUMAR R, SABARI K”** who carried out the project work under my supervision.

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ABSTRACT

Solar photovoltaic systems require continuous monitoring to maintain efficient power generation and identify abnormal operating conditions at an early stage. The proposed project develops an IoT-based solar power monitoring system using an ESP32 microcontroller and suitable voltage, current, temperature, and light sensors. The system continuously measures solar panel parameters and calculates the generated electrical power in real time. The collected information is transmitted through Wi-Fi to an IoT cloud platform where the user can monitor the solar system remotely. Fault conditions such as low voltage, low current, overheating, insufficient sunlight, and abnormal power generation are identified using predefined threshold values. When a fault is detected, the system provides an alert through indicators and IoT notifications so that corrective action can be taken quickly. The proposed system improves solar plant reliability, reduces manual inspection, supports preventive maintenance, and provides a low-cost solution for smart renewable-energy monitoring.

Keywords: Solar Power, Photovoltaic System, IoT, ESP32, Power Monitoring, Fault Detection, Voltage Sensor, Current Sensor, Temperature Monitoring, Real-Time Monitoring, Blynk, Renewable Energy.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	MEANING
ESP32	Espressif32-bit Microcontroller
IoT	Internet of Things
DC	Direct Current
AC	Alternating Current
V	Voltage
A	Current
W	Watt

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF SOLAR ENERGY

Solar energy is one of the most important renewable energy sources because sunlight is freely available and can be converted directly into electrical energy using photovoltaic panels. Solar photovoltaic systems are increasingly used in homes, industries, agricultural applications, educational institutions, and large-scale solar farms. The growing demand for clean energy has increased the importance of reliable solar power generation and efficient monitoring systems.

A solar photovoltaic system consists of solar panels, a charge controller, battery or electrical load, inverter where required, and associated electrical connections. The solar panel converts sunlight into electrical energy, while the remaining components regulate, store, or utilize the generated power. The amount of electricity produced depends on factors such as solar radiation, temperature, panel condition, wiring, and load.

Although solar systems require relatively low maintenance, their performance can decrease due to dust accumulation, shading, high temperature, damaged cells, loose connections, sensor problems, and electrical faults. If these conditions are not identified at an early stage, the generated power can decrease and the lifetime of the equipment can be affected. Therefore, continuous monitoring is important for maintaining reliable solar power generation.

The proposed project uses IoT technology to continuously monitor solar power generation and identify abnormal operating conditions. The ESP32 microcontroller collects sensor data, processes the measured parameters, and sends the information

through Wi-Fi to an IoT platform. The system provides real-time information to the user and generates alerts when predefined fault conditions are detected.

1.2 SOLAR PHOTOVOLTAIC SYSTEMS

Solar photovoltaic systems operate by converting solar radiation into direct-current electrical power. The electrical output of a solar panel depends mainly on the intensity of sunlight and the characteristics of the photovoltaic cells. The generated voltage and current can be measured to determine the instantaneous power output of the panel.

In a basic solar system, the panel is connected to a charge controller, battery, and load. The charge controller regulates the charging process and protects the battery. In larger installations, an inverter may convert DC power into AC power for household or industrial loads.

The output of a photovoltaic panel is not constant throughout the day. Morning, noon, evening, cloud cover, and shading can produce different levels of generation. Consequently, a monitoring system must continuously measure the electrical and environmental parameters rather than depending on a single measurement.

The proposed project addresses this requirement by measuring voltage, current, temperature, and light intensity. These measurements allow the controller to calculate generated power and identify conditions that may indicate a fault or reduction in performance.

1.3 CHALLENGES IN SOLAR POWER MONITORING

Conventional solar installations may use meters that display electrical parameters locally. Such systems require the operator to physically inspect the equipment. This becomes difficult when the solar installation is located on a rooftop,

remote area, agricultural field, or large solar farm.

Another major challenge is delayed fault identification. A panel may generate less power because of shading, dust, damaged cells, or wiring problems. If the operator does not notice the reduction immediately, energy losses can continue for a long period.

Temperature also affects solar panel performance. Excessive temperature can reduce electrical performance and may indicate inadequate ventilation or abnormal operating conditions. Battery systems can also experience overcharging or under-voltage conditions if they are not properly monitored.

IoT technology provides a practical solution to these problems by allowing measurements to be collected automatically and transmitted to a remote monitoring platform. The proposed system therefore reduces dependence on manual inspection and provides faster information about abnormal conditions.

1.4 NEED FOR INTELLIGENT MONITORING

Intelligent monitoring systems are required because solar power generation is affected by changing environmental and electrical conditions. Monitoring only one parameter is not sufficient to determine whether the system is operating normally.

By measuring voltage and current simultaneously, the system can calculate instantaneous power. Temperature and light measurements provide additional information that can help explain variations in generated power.

An IoT platform further improves monitoring by allowing the user to view the system remotely. Historical measurements can also be used to understand generation patterns and identify unusual changes.

Therefore, the combination of sensors, ESP32 processing, Wi-Fi communication, cloud monitoring, and fault alerts provides a more effective solution.

1.5 PROPOSED SYSTEM OVERVIEW

The proposed system consists of a solar panel, sensors, ESP32 microcontroller, battery/charge-control section, alert devices, and IoT platform. The sensors continuously measure the operating parameters of the solar system.

The ESP32 receives sensor values and processes them. Voltage and current measurements are used to calculate power using the relationship between voltage and current. Temperature and light readings are used to provide additional operating information.

The controller compares measured values with predefined limits. If a parameter moves outside the acceptable range, the system classifies the condition as an abnormal condition or fault.

The processed information is transmitted through Wi-Fi to the IoT dashboard. The user can therefore observe the solar system remotely and receive notifications when faults are detected.

1.6 SCOPE OF THE PROJECT

The project focuses on real-time monitoring and basic fault detection for a small solar photovoltaic system. The system is designed around an ESP32 microcontroller and commonly available sensors.

The monitoring parameters include solar voltage, solar current, calculated power, temperature, light intensity, and battery voltage where applicable. These parameters provide a useful representation of the operating condition of the system.

The project also includes IoT-based visualization. Sensor values can be displayed on a mobile or web dashboard, allowing the user to observe system performance without being physically present near the solar panel.

The system can later be expanded for larger solar installations, multiple panels, predictive maintenance, artificial intelligence, weather integration, and automated cleaning systems.

1.7 OBJECTIVES OF THE PROJECT

The main objective is to develop an IoT-based solar power monitoring system capable of continuously measuring important operating parameters. The system should provide reliable real-time information about power generation.

A second objective is to detect abnormal conditions automatically. Fault conditions can include low voltage, abnormal current, high temperature, insufficient light, and battery-related conditions.

Another objective is to transmit sensor data through Wi-Fi to an IoT platform. This enables remote monitoring and reduces the need for continuous physical inspection.

The final objective is to develop a low-cost, simple, reliable, and expandable prototype that can serve as a foundation for advanced solar monitoring and predictive maintenance systems.

1.8 NEED FOR THE PROJECT

Solar systems can lose generation because of environmental conditions and equipment faults. Early identification of these conditions can help reduce energy losses and maintenance delays.

Manual inspection is particularly inconvenient for remote installations. An automated monitoring system can continuously observe the parameters and provide information whenever a problem occurs.

IoT connectivity allows the user to monitor the system from a mobile phone or computer. This improves accessibility and makes the system suitable for distributed solar installations.

Thus, the project is needed to improve solar system visibility, reduce fault-detection time, and support efficient operation and maintenance.

CHAPTER 2

LITERATURE REVIEW

2.1 SOLAR PV MONITORING SYSTEMS

Solar monitoring systems are developed to observe electrical and environmental parameters associated with photovoltaic generation. Typical measurements include voltage, current, power, temperature, and solar radiation.

Traditional systems often depend on locally installed meters. Such meters can provide useful information but may not support remote monitoring or automatic notifications.

Microcontrollers have made it possible to develop compact monitoring systems that collect measurements from multiple sensors. These systems can process data and calculate power automatically.

The proposed project builds upon this concept by combining solar measurement with IoT communication and automatic fault identification.

2.2 IoT-BASED SOLAR MONITORING

IoT technology connects physical devices to communication networks and cloud platforms. In a solar monitoring application, sensor data can be collected by a microcontroller and transmitted through Wi-Fi.

An IoT dashboard can display the measurements in numerical and graphical form. This allows users to observe changes in power generation over time. Remote monitoring is particularly useful for rooftop and remote solar systems because the operator does not need to be physically present near the installation.

The proposed system uses the ESP32's wireless capability to provide this connectivity and make solar system information accessible remotely.

2.3 SOLAR FAULT DETECTION

Solar PV faults can appear as abnormal electrical measurements. For example, an unexpected reduction in current may indicate shading, wiring problems, or panel damage.

High temperature may indicate abnormal environmental or operating conditions. Voltage outside the expected range may indicate electrical or connection problems.

Fault detection can be performed by comparing measured parameters with predefined thresholds. This approach is simple and suitable for a student-level prototype.

More advanced systems can use historical data and machine-learning techniques. However, the proposed project focuses primarily on threshold-based detection with IoT monitoring.

2.4 REMOTE MONITORING SYSTEMS

Remote monitoring reduces the requirement for physical inspection. Sensor data can be transmitted to an online platform and accessed through a mobile phone or computer.

The user can observe the current operating state and identify changes in generation. Alerts can also be generated when a predefined condition occurs.

Such a system is useful when the solar installation is located far from the operator. It can also help maintenance personnel prioritize inspections.

Therefore, remote monitoring improves the practical usefulness of a solar power monitoring system.

2.5 SENSOR – BASED MONITORING

Sensors provide the basic data required for intelligent monitoring. A voltage/current sensor measures electrical output, while a temperature sensor measures the thermal condition of the system.

An LDR can be used to estimate relative light intensity. Comparing light intensity with electrical generation can help identify conditions such as insufficient sunlight.

The ESP32 collects the sensor values and processes them. Proper sensor calibration is important to obtain meaningful measurements.

The proposed system combines multiple sensors so that the condition of the solar system can be evaluated using more than one parameter.

2.6 RESEARCH GAP

Many basic monitoring systems display voltage and current but do not provide integrated fault detection. Other systems may focus on IoT communication without detailed fault classification.

A useful system should combine measurement, processing, fault identification, and remote notification. This integration can reduce the delay between fault occurrence and user awareness.

The proposed project addresses this requirement by combining solar sensors with ESP32 processing and IoT connectivity.

The research gap therefore provides the motivation for developing a simple, low-cost solar monitoring and fault-detection prototype.

CHAPTER 3

PROBLEM IDENTIFICATION

3.1 PROBLEM IDENTIFIED

Solar panels are exposed to environmental conditions continuously. Dust, shading, high temperature, and weather changes can affect the amount of power generated.

Electrical problems can also occur because of loose connections, damaged cables, abnormal voltage, or reduced current. These faults may not always be visible through physical inspection.

Without continuous monitoring, the operator may only discover the problem after noticing a reduction in electricity generation.

Therefore, a system capable of continuously measuring solar parameters and generating fault alerts is required.

3.1.1 LOW POWER GENERATION

Low power generation can occur when sunlight intensity decreases or when the panel is partially shaded. It may also occur because of dust or physical damage.

A monitoring system can compare the measured power with an expected or predefined value.

If the power remains significantly low under adequate light conditions, the system can generate a warning.

This helps the operator investigate the panel, wiring, or environmental conditions.

3.1.2 PANEL OVERHEATING

Solar panels operate outdoors and can reach high temperatures. Excessive temperature can affect their electrical performance.

The temperature sensor provides information about the panel or surrounding operating temperature.

When the temperature exceeds a predefined threshold, the controller can classify the condition as an over-temperature condition.

The user can then inspect ventilation, mounting conditions, and panel operation.

3.1.3 VOLTAGE ABNORMALITY

The generated solar voltage changes depending on sunlight and load conditions. An unusually low voltage may indicate a connection or panel problem.

An unusually high voltage may be unsafe for connected equipment depending on the system design.

The voltage sensor provides continuous measurement to the ESP32.

The controller compares the measured value with the defined operating range and generates an alert if required.

3.1.4 CURRENT ABNORMALITY

Current is directly related to the amount of power being delivered under a given voltage. A sudden decrease in current can occur due to shading, dirty panels, open circuits, or connection problems.

The current sensor continuously monitors the electrical output. The system uses this information together with voltage and light measurements to identify abnormal operation.

3.1.5 LACK OF REAL-TIME MONITORING

Manual monitoring cannot provide continuous information about a solar system. An operator may inspect the system only at selected intervals. Faults that occur between inspections can remain undetected for a significant period.

IoT monitoring solves this problem by sending data continuously to a remote dashboard. The proposed system therefore provides a real-time view of solar power generation.

3.1.6 DELAYED FAULT DETECTION

Delayed fault detection can result in energy loss and unnecessary equipment downtime. A fault alert system can identify abnormal conditions as soon as sensor readings cross predefined thresholds.

The ESP32 processes measurements locally, allowing rapid fault identification. Remote notification further reduces the time required for the operator to become aware of the problem.

3.1.7 NEED FOR INTELLIGENT SOLAR MONITORING

A solar monitoring system should not only display values but should also interpret them. Combining voltage, current, temperature, and light measurements provides more information than monitoring one parameter alone.

The proposed system uses predefined rules to classify abnormal operating conditions. This provides a foundation for future predictive maintenance and AI-based fault detection.

3.2 PROPOSED SOLUTION

The proposed solution uses an ESP32 microcontroller as the central control unit. Voltage and current measurements are obtained from an INA219 sensor, while

temperature is measured using a DHT22 sensor. An LDR provides information about relative light intensity. The controller calculates solar power using measured voltage and current.

The measured parameters are compared with predefined thresholds. When an abnormal condition is detected, the system activates local warning devices and transmits the status through the IoT platform. The complete solution therefore combines sensing, processing, fault detection, wireless communication, cloud monitoring, and alerts.

3.3 OBJECTIVES

The system aims to continuously monitor solar power generation.

- It aims to detect electrical and environmental abnormalities.
- It aims to provide real-time IoT-based remote monitoring.
- It aims to generate immediate fault indications.
- It is also designed to be low-cost, simple, expandable, and suitable for future smart solar applications.

CHAPTER 4

METHODOLOGY

4.1 PROJECT METHODOLOGY

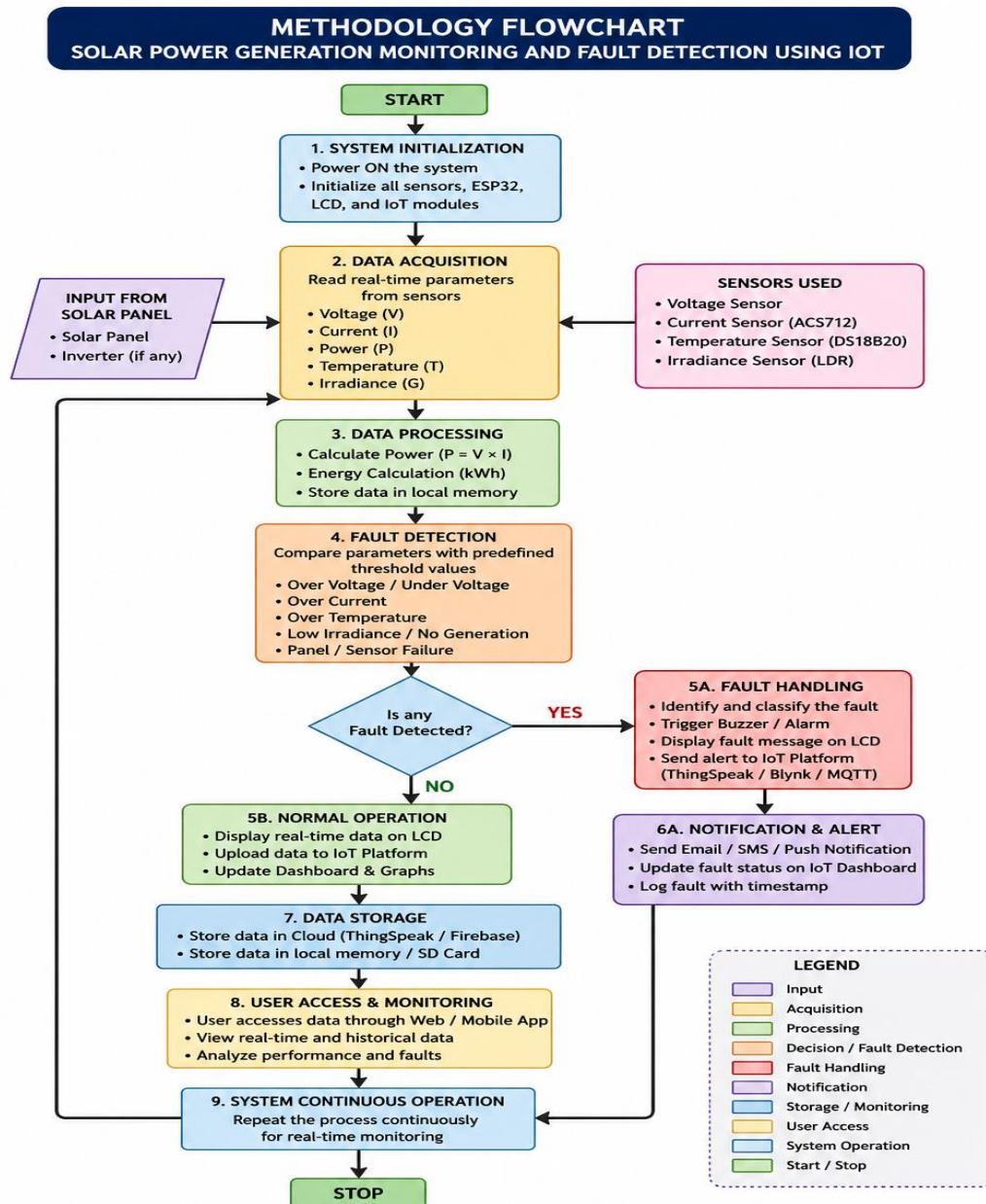


FIG 4.1 METHODOLOGY FLOWCHART

The methodology begins with the solar panel generating electrical energy from sunlight. Sensors are connected to measure important electrical and environmental parameters. The INA219 sensor measures voltage and current. The DHT22 measures temperature, while the LDR provides an indication of light intensity.

The ESP32 receives these sensor values and processes them. It calculates the generated power and evaluates the measured parameters against predefined thresholds. The processed information is transmitted through Wi-Fi to an IoT platform, where the user can view the data and receive alerts.

4.2 DATA ACQUISITION

Data acquisition is the first important stage of the system. Sensors continuously collect information from the solar power system. Voltage and current measurements provide the electrical output. Temperature and light measurements provide environmental information.

The ESP32 reads the sensor values at regular intervals. Continuous data acquisition enables the system to identify changes in system performance.

4.3 SENSOR MEASUREMENT

The voltage/current sensor provides electrical measurements to the controller through digital communication. The DHT22 provides temperature information, while the LDR provides a light-dependent signal.

Each sensor is selected according to its suitability for the prototype. The controller combines all sensor information before performing fault analysis.

4.4 ESP32 PROCESSING

The ESP32 acts as the main control unit. It receives sensor readings and performs the required calculations.

The instantaneous solar power is calculated using:

$$\mathbf{P} = \mathbf{V} \times \mathbf{I}$$

where **P** is power in watts, **V** is voltage in volts, and **I** is current in amperes.

The calculated and measured values are then compared with the predefined operating limits.

4.5 IoT COMMUNICATION

- The ESP32 uses its Wi-Fi capability to transmit the collected data.
- The IoT platform receives the data and displays it on a dashboard.
- The dashboard can contain widgets for voltage, current, power, temperature, and fault status.
- This allows the user to monitor the system remotely.

4.6 FAULT DETECTION

Fault detection is performed by comparing the measured parameters against threshold values.

For example, if the measured temperature exceeds the defined maximum operating temperature, an over-temperature condition is generated.

Similarly, low voltage or abnormal current can produce corresponding fault conditions. The threshold values can be adjusted according to the selected solar panel and system configuration.

4.7 ALERT GENERATION

When a fault is detected, the ESP32 activates a local warning mechanism. A buzzer can provide an audible warning, while LEDs can indicate the type or severity of the condition.

The IoT platform can also generate a remote notification. The combination of

local and remote alerts improves the effectiveness of the monitoring system.

4.8 CLOUD MONITORING

- The IoT dashboard displays current measurements and system status.
- The user can observe the change in power generation and identify abnormal values.
- Historical information can help understand the performance of the solar system over time.
- Cloud monitoring therefore provides a convenient method for remote supervision.

CHAPTER 5

COMPONENT SELECTION

5.1 ESP32 DEVELOPMENT BOARD



FIG 5.1 ESP32 DEVELOPMENT BOARD

The ESP32 Development Board is selected as the main controller of the Solar Power Generation Monitoring and Fault Detection System. It is a low-cost microcontroller with built-in Wi-Fi and Bluetooth capabilities. The ESP32 receives data from the voltage, current, temperature, and light sensors connected to the solar system. It processes the collected data and calculates important parameters such as generated power.

The ESP32 also performs the fault-detection operation by comparing sensor readings with predefined threshold values. When an abnormal condition is detected, it controls the buzzer and LED indicators and sends the fault information to the IoT platform through Wi-Fi. Its multiple GPIO pins, ADC capability, I²C communication, and wireless connectivity make it suitable as the central controller of the proposed project.

5.2 SOLAR PANEL



FIG 5.2 SOLAR PANEL

The 20 W solar panel is the main power-generation component of the proposed system. It converts solar radiation into electrical energy through photovoltaic cells. The electrical output of the panel varies depending on sunlight intensity, temperature, shading, dust, and other environmental conditions. The generated voltage and current are continuously monitored by the system.

The solar panel supplies electrical energy to the connected solar-power system and battery charging section. Its voltage and current are measured using the INA219 sensor, and the ESP32 calculates the generated power using the equation $P = V \times I$. Monitoring the panel output allows the system to identify reductions in generation and possible abnormal operating conditions.

5.3 INA219 VOLTAGE AND CURRENT SENSOR

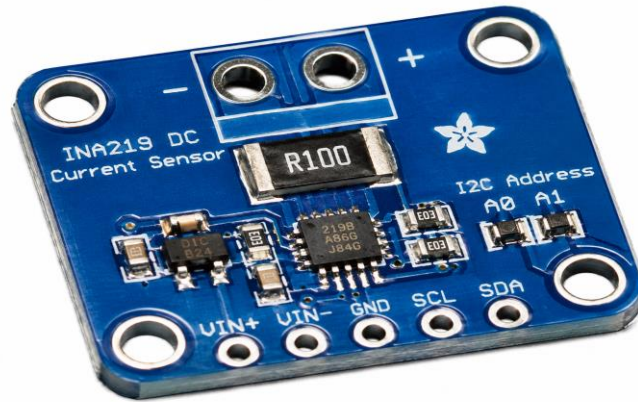


FIG 5.3 INA219 VOLTAGE AND CURRENT SENSOR

The INA219 voltage and current sensor is used to measure the electrical parameters produced by the solar panel. It can measure voltage and current and provide the measured information digitally to the ESP32. The sensor communicates with the controller through the I²C interface, using SDA and SCL connections.

The measured voltage and current values are used to calculate the instantaneous solar power. The ESP32 continuously checks these values against predefined limits to identify conditions such as low voltage, abnormal current, or reduced

power generation. Therefore, the INA219 is an important component for both real-time monitoring and fault detection.

5.4 DHT22 TEMPERATURE SENSOR

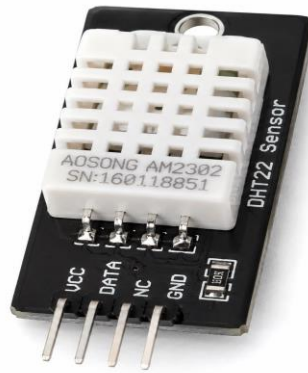


FIG 5.4 DHT22 TEMPERATURE SENSOR

The DHT22 temperature sensor is used to measure the temperature associated with the solar monitoring system. Temperature is an important parameter because changes in temperature can affect photovoltaic performance. The DHT22 provides digital temperature data to the ESP32, allowing the controller to monitor temperature continuously.

The measured temperature is displayed on the IoT dashboard along with the solar voltage, current, and power. If the temperature exceeds a predefined threshold, the ESP32 can identify an over-temperature condition and activate the buzzer and fault LED. The temperature information can also be stored for future analysis of solar-panel performance.

5.5 LDR LIGHT SENSOR

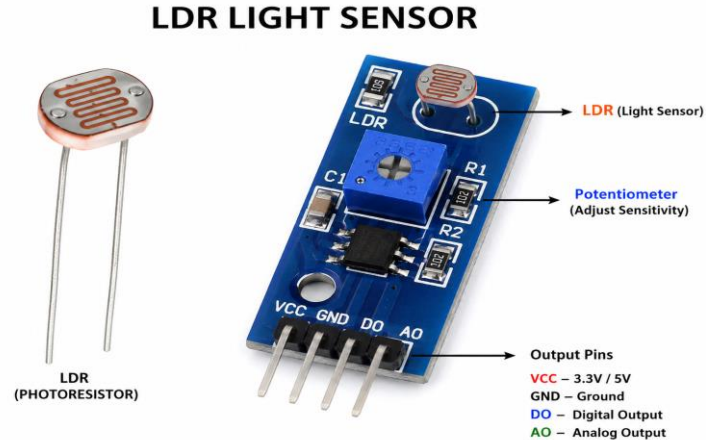


FIG 5.5 LDR LIGHT SENSOR

The LDR (Light Dependent Resistor) is used to detect the relative intensity of light falling on or around the solar panel. The resistance of the LDR changes according to the amount of incident light. The LDR module provides an electrical signal that can be read by an analog input of the ESP32.

The light-intensity information helps the system understand changes in solar power generation. For example, low sunlight naturally results in lower power generation, whereas high light intensity with unexpectedly low power may indicate a possible problem with the panel or electrical connection. Thus, the LDR provides useful environmental information for fault analysis.

5.6 SOLAR CHARGE CONTROLLER



FIG 5.6 SOLAR CHARGE CONTROLLER

The solar charge controller is used to control the electrical energy supplied from the solar panel to the rechargeable battery. It regulates the charging process and helps prevent inappropriate charging conditions. The charge controller is normally connected between the solar panel and battery in the proposed solar power system.

The charge controller improves the safety and reliability of the battery-storage section. Battery-related information can also be monitored by the ESP32 to identify low-voltage or abnormal charging conditions. The selected charge controller should be compatible with the voltage, current, panel rating, and battery type used in the project.

5.7 12 V RECHARGEABLE BATTERY



FIG 5.7 12 V RECHARGEABLE BATTERY

The 12 V rechargeable battery is used to store electrical energy generated by the solar panel. The stored energy can be used when solar radiation is insufficient or when the solar panel is not generating power. The battery therefore helps maintain the availability of electrical energy in the system.

Battery voltage can be monitored as an additional parameter of the proposed IoT system. If the battery voltage falls below a predefined value, the ESP32 can generate a low-battery warning. The battery must be connected through a suitable solar charge

controller to ensure proper charging and protection.

5.8 1-CHANNEL RELAY MODULE



FIG 5.8 1-CHANNEL RELAY MODULE

The 1-channel relay module is used for automatic switching of an electrical load. The relay can be controlled by an output signal from the ESP32. It provides a convenient method for turning a connected load ON or OFF according to the condition of the solar system.

The relay can be used as an optional protection and control mechanism in the project. For example, the ESP32 can disconnect a selected load during a critical battery condition or abnormal operating condition. The relay module should be selected according to the voltage and current requirements of the controlled load.

5.9 BUZZER



FIG 5.9 BUZZER

The buzzer is used as an audible fault indicator in the solar monitoring system. It provides an immediate warning when the ESP32 detects an abnormal condition. The buzzer can be activated for conditions such as high temperature, abnormal voltage, low current, low power generation, or battery problems.

The buzzer is controlled by the ESP32 through a digital output. It provides a local warning even when the user is not viewing the IoT dashboard. This makes the fault-detection system more effective because both local and remote notifications can be provided.

5.10 LED INDICATORS

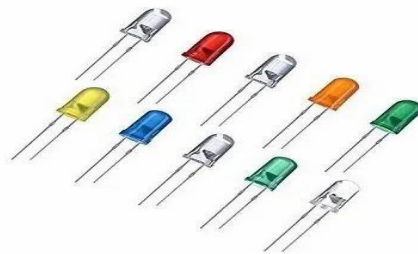


FIG 5.10 LED INDICATORS

LED indicators are used to show the operating condition of the solar monitoring system. A green LED can be used to indicate normal operation, while a red LED can indicate the presence of a fault. The LEDs are connected to ESP32 GPIO pins through suitable current-limiting resistors.

During normal operation, the green LED can remain ON to indicate that the system is functioning correctly. When a fault is detected, the red LED can be activated by the ESP32. LED indicators provide a simple visual indication and are especially useful during prototype testing and demonstration.

5.11 JUMPER WIRES



FIG 5.12 JUMPER WIRES

Jumper wires are used to establish electrical connections between the ESP32, sensors, LEDs, buzzer, and other components. They provide connections for power, ground, digital signals, analog signals, and communication interfaces such as I²C.

Different types of jumper wires, such as male-to-male, male-to-female, and female-to-female wires, can be used according to the connector requirements. They make the prototype easy to assemble and modify during testing. Proper wiring and common grounding are important for reliable sensor operation.

5.12 220 Ω RESISTORS



FIG 5.13 220 Ω RESISTORS

The 220 Ω resistors are used mainly as current-limiting resistors for the LED indicators. When an LED is connected directly to a microcontroller output without current limiting, excessive current may flow through the LED and damage the LED or the controller output.

The resistor is therefore connected in series with each LED to limit the current to a suitable value. In the proposed system, one resistor can be used with the green LED and another with the red LED. This simple protection component helps ensure reliable operation of the visual fault-indication circuit.

5.14 5 V REGULATED POWER SUPPLY



FIG 5.14 5V REGULATED POWER SUPPLY

The 5 V regulated power supply is used to provide stable electrical power to the ESP32 and associated electronic components during prototype operation. A regulated supply is important because unstable voltage can cause microcontroller resets, incorrect sensor readings, or unreliable communication.

The power supply can be provided using a suitable USB adapter or regulated DC supply. The supply must have adequate current capacity for the ESP32, sensors, LEDs, and buzzer. In the final solar-powered version, a suitable DC-DC regulator can be used to obtain the required voltage from the battery system.

5.15 Wi-Fi NETWORK

The Wi-Fi network provides wireless communication between the ESP32 and the IoT platform. Since the ESP32 contains integrated Wi-Fi functionality, an additional wireless communication module is not required. The controller connects to the available network and transmits sensor data through the Internet.

Wi-Fi enables remote monitoring of solar voltage, current, power, temperature, light intensity, and fault status. When a fault is detected, the ESP32 can send the corresponding information to the IoT dashboard. The availability and stability of the Wi-Fi connection therefore influence the remote-monitoring function of the project

5.16 IoT PLATFORM / BLYNK

The Blynk IoT platform can be used to display and monitor the data collected by the ESP32. The platform provides a dashboard where parameters such as voltage, current, power, temperature, and battery voltage can be displayed in real time. Graphs and status indicators can also be used to observe changes in solar generation.

The IoT platform also supports remote fault notification. When the ESP32 detects an abnormal condition, the fault status can be transmitted through Wi-Fi and displayed to the user. This enables monitoring of the solar system from a mobile device or computer without requiring physical access to the solar installation.

CHAPTER 6

SYSTEM OPERATION

6.1 SOLAR POWER GENERATION

The operation of the proposed system begins with the solar panel, which converts sunlight into electrical energy. The amount of electrical energy generated depends mainly on solar irradiance and environmental conditions. The generated DC voltage and current are supplied to the monitoring and power-management section of the system.

The electrical output of the solar panel is continuously measured using the INA219 voltage and current sensor. The sensor provides the measured values to the ESP32 through the I²C communication interface. At the same time, the DHT22 measures temperature and the LDR provides an indication of the available light intensity.

The ESP32 receives all these measurements and processes them to determine the operating condition of the solar panel. The generated power is calculated from the measured voltage and current using $P = V \times I$. These parameters form the basic input for real-time solar-power monitoring and subsequent fault detection.

6.2 DATA ACQUISITION AND PROCESSING

The ESP32 acts as the central controller of the system. It collects voltage and current information from the INA219, temperature information from the DHT22, and light-intensity information from the LDR. The controller periodically reads these parameters so that the condition of the solar system can be monitored continuously.

After receiving the sensor data, the ESP32 processes the readings and calculates the instantaneous generated power. The measured parameters are compared

with predefined threshold values. This allows the controller to distinguish between normal operating conditions and abnormal conditions.

The processed information is then prepared for both local indication and IoT transmission. During normal operation, the system indicates the healthy condition using the appropriate LED indication. The measured values are also transmitted through Wi-Fi so that they can be viewed remotely on the IoT dashboard.

6.3 FAULT DETECTION AND IDENTIFICATION

The fault-detection section continuously compares the measured parameters with their predefined acceptable ranges. Abnormal voltage, unusually low current, reduced power generation, excessive temperature, or insufficient light conditions can be identified by the controller. The measured parameters are considered together rather than depending on only one sensor.

When the ESP32 detects a parameter outside its expected range, it determines that an abnormal condition has occurred. The system can distinguish conditions such as low voltage, abnormal current, high temperature, low light intensity, or reduced solar power generation according to the programmed threshold values.

After detecting a fault, the ESP32 activates the local warning devices. The fault LED and buzzer provide an immediate indication to the user. At the same time, the detected fault condition is transmitted through Wi-Fi to the IoT platform, allowing the user to monitor the problem remotely.

6.4 IoT MONITORING AND NOTIFICATION

The ESP32 uses its built-in Wi-Fi communication to transmit the measured solar-system data to the IoT platform. Parameters such as voltage, current, calculated power, temperature, light intensity, and fault status can be displayed on the

dashboard. This provides real-time visibility of the solar system.

The IoT dashboard allows the user to monitor the system without being physically present near the solar panel. Graphs and numerical indicators can be used to observe changes in solar generation over time. When an abnormal condition is detected, the corresponding fault status can also be displayed to the user.

The complete operation is therefore continuous: sensor data acquisition → processing → power calculation → threshold comparison → fault detection → local indication → IoT transmission → dashboard monitoring. After completing one monitoring cycle, the ESP32 returns to sensor acquisition and repeats the process continuously.

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 HARDWARE IMPLEMENTATION

Hardware implementation begins by assembling the ESP32 and the selected sensors. INA219 is connected to the solar electrical measurement path, while DHT22 and LDR are connected for environmental sensing. LEDs and a buzzer are connected to suitable GPIO pins. An optional relay can also be included for load control.

The solar panel and supporting power components are connected according to the designed circuit. If a battery is used, a suitable solar charge controller is placed between the panel and battery. All connections must be checked for correct polarity and voltage levels. The components should be securely mounted to prevent loose connections.

Each hardware component should be tested individually before complete integration. Sensor communication is checked using the ESP32, and the output indicators are tested separately. After individual testing, all components are connected together. The complete system is then powered and tested under controlled conditions.

7.2 SOFTWARE IMPLEMENTATION

The software implementation begins by initializing the ESP32 and connected peripherals. The program configures the sensor interfaces, GPIO pins, and I²C communication. It then establishes the Wi-Fi connection required for IoT communication. After successful initialization, the monitoring process begins.

During each program cycle, the ESP32 reads voltage, current, temperature, and light intensity. It calculates generated power from the electrical measurements.

The program then compares the values with predefined threshold limits. The resulting system status determines the operation of the LEDs, buzzer, and optional relay.

After processing the data, the ESP32 sends the values to the IoT dashboard. The dashboard is updated with the latest measurements and fault status. The program then waits for the next monitoring cycle. This process continues as long as the system remains powered and connected.

7.3 SENSOR INTEGRATION

Sensor integration combines all the measurement devices into a single ESP32-based system. INA219 communicates using I²C, while DHT22 uses a digital data connection. The LDR is connected to an analog input. These different sensor interfaces can be handled by the ESP32.

During operation, the controller reads each sensor and stores the measured values for processing. The measurements are combined to provide a complete view of the solar system. Voltage and current are used to calculate power, while temperature and light intensity provide environmental information. This integrated data improves monitoring capability.

The combined sensor values are also transmitted to the IoT dashboard. The user can compare electrical output with environmental conditions. This makes it easier to understand whether changes in power are related to sunlight or another condition. Sensor integration therefore supports both monitoring and fault detection.

7.4 IoT DASHBOARD

The IoT dashboard provides the user interface for remote monitoring. The ESP32 sends measured parameters to the dashboard through a Wi-Fi connection. The dashboard receives the information and displays the latest values. This provides convenient access to solar-system data.

The main dashboard parameters can include voltage, current, power, temperature, and light intensity. A separate indicator can display normal or fault operation. Graphs can be used to observe parameter changes over time. The interface should be arranged clearly so that important information is easy to identify.

The dashboard can also support fault monitoring. When the ESP32 detects an abnormal condition, it sends the corresponding fault status. The user can observe this information remotely and take appropriate action. This makes the IoT dashboard an important part of the complete monitoring system.

CHAPTER 8

METHODOLOGY AND ALGORITHM

8.1 METHODOLOGY

The proposed methodology begins with solar-power generation and sensor-based data collection. The solar panel generates electrical power, while INA219 measures voltage and current. DHT22 and LDR measure temperature and light intensity. The ESP32 collects all these parameters for processing.

The controller calculates the generated power using the measured voltage and current. It then compares the electrical and environmental parameters with predefined threshold values. Based on the comparison, the controller determines whether the system is operating normally. If an abnormal condition is detected, a fault status is generated.

The processed information is transmitted to the IoT dashboard through Wi-Fi. Normal conditions result in normal-status indication, while abnormal conditions activate the warning devices. The monitoring cycle is then repeated continuously. This methodology provides real-time monitoring and basic automatic fault detection.

8.2 FLOWCHART

The system starts with initialization of the ESP32, sensors, output devices, and communication interface. The controller establishes the Wi-Fi connection before beginning normal monitoring. Once initialization is complete, the ESP32 starts reading the sensor values. This begins the main monitoring loop.

The controller reads voltage, current, temperature, and light intensity. It calculates power using the measured voltage and current. The values are then compared with their respective threshold conditions. The controller determines whether the system is operating normally or whether a fault has occurred.

If the measurements remain within the acceptable limits, the normal-status

indication is maintained. If an abnormal condition is detected, the red LED and buzzer are activated and the fault status is transmitted. The system then returns to the measurement stage. This process continues repeatedly during operation.

8.3 FAULT DETECTION ALGORITHM

The fault-detection algorithm begins by acquiring the electrical and environmental parameters. Voltage and current are read from INA219, while temperature and light intensity are obtained from DHT22 and LDR. The ESP32 calculates the generated power from the measured voltage and current. These values become the input to the detection process.

The controller compares each parameter against its predefined threshold. If a parameter exceeds the permitted limit, the system identifies an abnormal condition. The controller can use multiple parameters to provide better interpretation of the operating condition. This reduces the possibility of interpreting every reduction in power as an electrical fault.

When a fault is detected, the ESP32 activates the warning outputs. The red LED and buzzer provide local notification, while the IoT dashboard displays the fault status remotely. When the measurements return to the acceptable range, the system can return to normal operation. The algorithm continuously repeats throughout the monitoring period.

8.4 IoT COMMUNICATION

IoT communication begins when the ESP32 connects to the available Wi-Fi network. The controller checks the connection before transmitting data. After communication is established, the sensor values are formatted into suitable data fields. These values are then sent to the selected IoT platform.

The dashboard receives the transmitted measurements and displays

them to the user. Voltage, current, power, temperature, and light can be updated periodically. The fault status is also transmitted as part of the monitoring data. This provides a remote view of the solar system.

Reliable communication is important for continuous IoT monitoring. Network interruptions can temporarily prevent dashboard updates. Local LEDs and the buzzer provide an additional indication when communication is unavailable or when a fault occurs. Future versions can use alternative communication methods to improve connectivity.

CHAPTER 9

TESTING AND RESULTS

9.1 SENSOR TESTING

Sensor testing begins with the INA219 electrical measurements. The sensor is connected to the solar output and its voltage and current values are observed. The ESP32 reads the measurements through the I²C interface. The values are checked to confirm that the electrical sensing section is operating correctly.

The DHT22 is tested by observing temperature measurements under different environmental conditions. The LDR is tested under different light levels to verify that its output changes accordingly. The ESP32 reads both sensor values and processes them. This confirms the basic operation of the environmental sensing section.

After individual testing, all sensors are integrated into the complete system. The ESP32 reads all parameters during the same monitoring cycle. The measurements are then processed and transmitted to the dashboard. Successful integration confirms that the sensor interfaces are working together correctly.

9.2 ELECTRICAL TESTING

Electrical testing is performed by observing the solar-panel voltage and current. The INA219 provides the measured electrical values to the ESP32. Different operating conditions can be used to observe changes in the output. These measurements are used to calculate generated power.

Power is calculated using the equation $P = V \times I$. For example, if the measured voltage is 18 V and current is 1 A, the calculated power is 18 W. The calculated value changes as the solar-panel output changes. This provides a useful indication of electrical performance.

The measured electrical values are also transmitted to the IoT dashboard. The user can observe voltage, current, and power remotely. The electrical information can be compared with light and temperature measurements. This

allows the overall solar-generation behavior to be analyzed.

9.3 FAULT TESTING

Fault testing is performed by creating controlled abnormal conditions. Reduced light can be used to test low-generation conditions, while selected electrical conditions can be used to test voltage or current thresholds. Temperature conditions can also be tested against the programmed limit. Each condition is observed to determine whether the system responds correctly.

When the measured value exceeds the selected threshold, the ESP32 generates a fault condition. The red LED is activated and the buzzer provides an audible warning. The fault status is also transmitted to the IoT dashboard. This allows both local and remote fault detection to be verified.

Testing must also confirm that normal conditions do not unnecessarily activate the alarm. The system should remain in normal status when all parameters are within their acceptable limits. Controlled testing helps verify the correctness of the programmed logic. The results can then be used to refine the threshold values.

9.4 IoT TESTING

IoT testing begins by connecting the ESP32 to the selected Wi-Fi network. The controller is checked for successful network communication. After connection, sensor measurements are transmitted to the IoT dashboard. The displayed values are then compared with the values obtained locally.

Voltage, current, power, temperature, and light values should appear correctly on the dashboard. The system status should also be updated according to the fault-detection logic. When a controlled fault is introduced, the dashboard should show the corresponding abnormal status. This verifies the remote-monitoring function.

The IoT test also evaluates the usefulness of remote access. The user can observe system information without being physically near the hardware. Fault information can be monitored through the dashboard. This demonstrates the main advantage of connecting the solar monitoring system to IoT.

CHAPTER 10

PERFORMANCE ANALYSIS

10.1 POWER ANALYSIS

Power analysis is performed using the voltage and current measured by the INA219 sensor. The ESP32 calculates instantaneous power by multiplying these two parameters. The calculated power changes according to solar radiation and electrical loading. Monitoring power therefore provides a direct indication of solar generation.

Higher sunlight generally provides greater energy availability to the photovoltaic panel. Reduced sunlight can result in lower electrical output. The LDR provides additional information about the available light condition. Comparing light intensity with power helps explain natural variations in generation.

Power values are transmitted to the IoT dashboard for remote observation. The user can observe changes in generation over time. Unexpected power reduction can be investigated using voltage, current, temperature, and light measurements. This provides a basic method of evaluating solar-system performance.

10.2 ENVIRONMENTAL ANALYSIS

Environmental analysis focuses mainly on temperature and light intensity. The DHT22 provides temperature measurements, while the LDR provides an indication of available light. Both parameters can change throughout the day. These changes can influence photovoltaic power generation.

Light intensity is particularly important because solar panels require sunlight to generate electricity. When light intensity decreases, the available solar energy generally decreases. Temperature can also affect photovoltaic behavior. Therefore, environmental measurements provide useful context for electrical output.

The environmental values are transmitted to the IoT dashboard along with the electrical parameters. The user can compare light, temperature, voltage, current, and power. This makes it easier to understand changes in the solar system. The collected information can also support future performance studies.

10.3 FAULT DETECTION ANALYSIS

Fault detection performance depends on the threshold values programmed into the ESP32. The controller continuously evaluates voltage, current, temperature, power, and other selected parameters. When a value moves outside its permitted range, the system generates a fault indication. This provides basic automated detection.

Multiple measurements can be considered together to improve interpretation. For example, low power under low light may be expected, while low power under adequate light may require inspection. Similarly, abnormal current with unusual voltage can indicate an electrical problem. Combining parameters therefore provides more useful information.

Fault information is provided through both local and remote methods. The red LED and buzzer indicate the condition at the hardware level. The IoT dashboard provides the same information remotely. This combination improves the overall usefulness of the fault-detection system.

10.4 IoT PERFORMANCE

IoT performance depends mainly on reliable Wi-Fi communication and correct data transmission. The ESP32 periodically sends the measured parameters to the IoT platform. The dashboard receives the information and updates the displayed values. Stable communication allows the user to observe the solar system continuously.

When the network connection is temporarily unavailable, remote updates may be interrupted. However, local indicators can continue to provide information

about the system condition. The software can also be designed to reconnect automatically when communication is restored. This improves the practical reliability of the monitoring system.

The IoT dashboard provides an effective interface for remote monitoring. The user can observe electrical and environmental parameters without visiting the installation. Fault conditions can also be identified remotely. Therefore, IoT communication significantly improves the accessibility of solar-system information.

CHAPTER 11

ADVANTAGES, APPLICATIONS AND LIMITATIONS

11.1 ADVANTAGES

The proposed system provides continuous monitoring of solar-panel performance. It measures important electrical and environmental parameters automatically. This reduces the need for frequent manual measurements. The collected information provides a better understanding of solar-system operation.

IoT communication allows remote monitoring of the solar installation. The user can observe voltage, current, power, temperature, and light intensity from an IoT dashboard. Fault information can also be transmitted remotely. This makes the system particularly useful for installations that are difficult to inspect physically.

The system uses a relatively simple hardware structure. The ESP32 combines processing and wireless communication in a single controller. Local LEDs and a buzzer provide immediate fault indication. The design can also be expanded with additional sensors and advanced software features.

11.2 APPLICATIONS

The proposed system can be used for monitoring small residential solar installations. Home users can observe the performance of their photovoltaic panels remotely. The system can provide information about power generation and selected abnormal conditions. This can reduce the need for frequent physical inspection.

The system can also be used for educational and laboratory applications. Students can learn about solar power, sensors, microcontrollers, IoT communication, and fault detection. The project provides a practical demonstration of renewable-energy monitoring. It can also be used as a foundation for advanced academic projects.

Other possible applications include solar street lighting. Small commercial photovoltaic installations can also use similar monitoring concepts. Larger

installations would require additional sensors and industrial-grade equipment. The basic architecture can nevertheless be adapted to different applications.

11.3 LIMITATIONS

The proposed system uses threshold-based fault detection, which has some limitations. It can identify only abnormal conditions that are represented by the programmed thresholds. Complex faults may not be recognized correctly. More advanced diagnostic algorithms would be required for comprehensive fault classification.

IoT monitoring also depends on network availability. If the Wi-Fi connection is interrupted, the dashboard may not receive new measurements. Local LEDs and the buzzer can still provide nearby status information. Future versions could use alternative communication technologies to improve connectivity.

Sensor accuracy is another important limitation. Incorrect sensor readings can affect power calculations and fault decisions. The hardware must therefore be connected and calibrated appropriately. Industrial solar installations would also require additional protection, isolation, and appropriately rated components.

11.4 COST ANALYSIS

The main hardware cost is associated with the ESP32, solar panel, sensors, battery, and charge controller. Additional components include the relay, LEDs, buzzer, wires, connectors, resistors, and supporting circuitry. The total cost depends on the selected component ratings and quantities. A small prototype can generally be constructed using relatively economical components.

Using the ESP32 reduces the requirement for a separate Wi-Fi communication module. Low-cost sensors such as the LDR and DHT22 also help keep the prototype economical. The modular design allows components to be added only when required. This makes the system suitable for academic development.

For a larger practical installation, higher-rated electrical components would be required. Therefore, cost should be evaluated according to the required system size and functionality. The proposed prototype provides a reasonable starting point for developing a larger system.

CHAPTER 12

CONCLUSION AND FUTURE SCOPE

12.1 CONCLUSIONS

The proposed Solar Power Generation Monitoring and Fault Detection Using IoT system provides an integrated approach to monitoring photovoltaic performance. The ESP32 collects voltage, current, temperature, and light-intensity measurements from the connected sensors. It calculates generated power and evaluates the operating condition of the solar system. This provides continuous information about solar performance.

The system also provides basic automatic fault detection using predefined threshold conditions. When an abnormal condition occurs, the ESP32 activates the red LED and buzzer. The fault information is simultaneously transmitted to the IoT dashboard. This provides both local and remote awareness of possible problems.

The integration of solar monitoring with IoT improves the accessibility of system information. The user can observe the solar parameters without being physically present at the installation. The proposed design is simple, economical, and expandable. It can serve as a foundation for more advanced solar monitoring and predictive-maintenance systems.

12.2 ACHIEVEMENTS

The project successfully combines solar-power generation with sensor-based monitoring. The INA219 provides voltage and current measurements, while DHT22 and LDR provide environmental information. The ESP32 processes the measurements and calculates generated power. These functions provide the basic monitoring capability of the project.

The system also implements threshold-based fault detection. The controller evaluates the measured parameters and determines whether they are within acceptable limits. Local LEDs and a buzzer provide immediate indication of

abnormal conditions. This demonstrates the ability of the system to identify selected faults automatically.

Remote monitoring is achieved through ESP32 Wi-Fi communication. The IoT dashboard displays the collected parameters and fault status. The user can therefore observe the solar system from a remote location. These features together satisfy the main objectives of the proposed project.

12.3 FUTURE ENHANCEMENTS

Machine-learning methods can be incorporated into future versions of the system. Historical measurements can be analyzed to identify complex patterns associated with different solar faults. Machine-learning models could classify faults more accurately than simple threshold methods. This would provide a more advanced diagnostic capability.

Cloud-based data storage can also be introduced for long-term monitoring. Historical voltage, current, power, temperature, and light data can be stored and analyzed. Graphs and performance reports can be generated automatically. This would allow users to study solar generation over longer periods.

Additional sensors and communication features can further improve the system. Battery voltage, charging current, panel temperature, and load consumption can be monitored. Mobile notifications, GPS, predictive maintenance, and multi-panel monitoring can also be added. These improvements could transform the prototype into a more comprehensive smart solar-management system.

12.4 FINAL SUMMARY

The Solar Power Generation Monitoring and Fault Detection Using IoT project provides a practical method for monitoring photovoltaic-system performance. The ESP32 serves as the central processing and communication unit, while INA219, DHT22, and LDR provide important measurements. The controller

calculates power and evaluates the system using predefined fault conditions. This provides continuous information about the solar installation.

The combination of local indicators and IoT communication improves the usefulness of the system. LEDs and a buzzer provide immediate local warnings, while the IoT dashboard provides remote access to the same information. The user can observe electrical and environmental parameters and identify selected abnormal conditions. This reduces dependence on manual monitoring.

Overall, the proposed system demonstrates the integration of renewable energy, embedded systems, sensors, fault detection, and IoT technology. Its modular design makes it suitable for academic projects and small-scale solar applications. With advanced algorithms, cloud storage, additional sensors, and predictive maintenance features, the system can be developed further. The project therefore provides a strong foundation for future smart solar-energy monitoring systems.

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PROTOTYPE PHOTOGRAPHIC